Mini Project 1

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Domain: Data science

Student Performance Prediction System

This project predicts student academic performance based on factors like attendance, study hours, previous grades, and socio-economic background. Machine learning models such as Logistic Regression, Decision Trees are used.

Objective

The main objective of this project is to:

- Predict student academic performance (Pass/Fail or Grades)
- Identify weak students early
- Help teachers take corrective actions
- Improve overall academic outcomes using data-driven insights

Problem Statement

In traditional education systems:

- Teachers cannot easily track individual student performance trends
- Early identification of weak students is difficult
- Decisions are based on intuition rather than data

This project aims to automate performance prediction using historical student data.

Technologies Used

- Programming Language: Python

• Libraries:

o Pandas (Data Handling)

o NumPy (Numerical Computation)

o Matplotlib / Seaborn (Visualization)

o Scikit-learn (ML Models)

• Tools: o Jupyter Notebook / Google Colab o Excel (Dataset)

Dataset description

1. student_id: A unique identifier for each student (Numerical).
2. weekly_self_study_hours: The number of hours a student spends studying outside of class each week (Feature).
3. attendance_percentage: The percentage of classes attended by the student (Feature).
4. class_participation: A score (0–10) representing how active the student is during lessons (Feature).
5. total_score: The cumulative marks obtained by the student (Numeric outcome).
6. grade: The final letter grade (A, B, C, D, F). This is the Target Variable for our Machine Learning models.

System Architecture

1. Data Collection

2. Data Preprocessing

3. Feature Selection

4. Model Training

5. Model Testing

6. Prediction Output

7. Performance Evaluation

System Modules

1. Data Input Module

2. Data Processing Module

3. Model Training Module

4. Prediction Module

5. Result Visualization Module

Methodology:

Step 1: data collection

Collect student data from:

o School/College records

o Sample datasets (CSV/Excel)

DATA INPUT MODULE

data collection

[1]:
```
import pandas as pd
df = pd.read_csv('student_performance.csv')
```

[2]:
```
print(df.head())
```

```
   student_id  weekly_self_study_hours  attendance_percentage  \
0           1                     18.5                   95.6
1           2                     14.0                   80.0
2           3                     19.5                   86.3
3           4                     25.7                   70.2
4           5                     13.4                   81.9

   class_participation  total_score grade
0                  3.8         97.9     A
1                  2.5         83.9     B
2                  5.3        100.0     A
3                  7.0        100.0     A
4                  6.9         92.0     A
```

Step 2 : data processing

DATA PROCESSING MODULE

data preprocessing

```
[29]: #convert categorical data into numerical data A is the best grade F is bad grade
grade_map = {'A': 0, 'B': 1, 'C': 2, 'D': 3, 'F': 4}    #dictionary
```

```
[4]: df.isnull()
```

[4]:

	student_id	weekly_self_study_hours	attendance_percentage	class_participation	total_score	grade
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
999995	False	False	False	False	False	False
999996	False	False	False	False	False	False
999997	False	False	False	False	False	False
999998	False	False	False	False	False	False
999999	False	False	False	False	False	False

1000000 rows × 6 columns

```
[5]: df['grade_num'] = df['grade'].map(grade_map)
```

```
[6]: print(df['grade_num'])
```

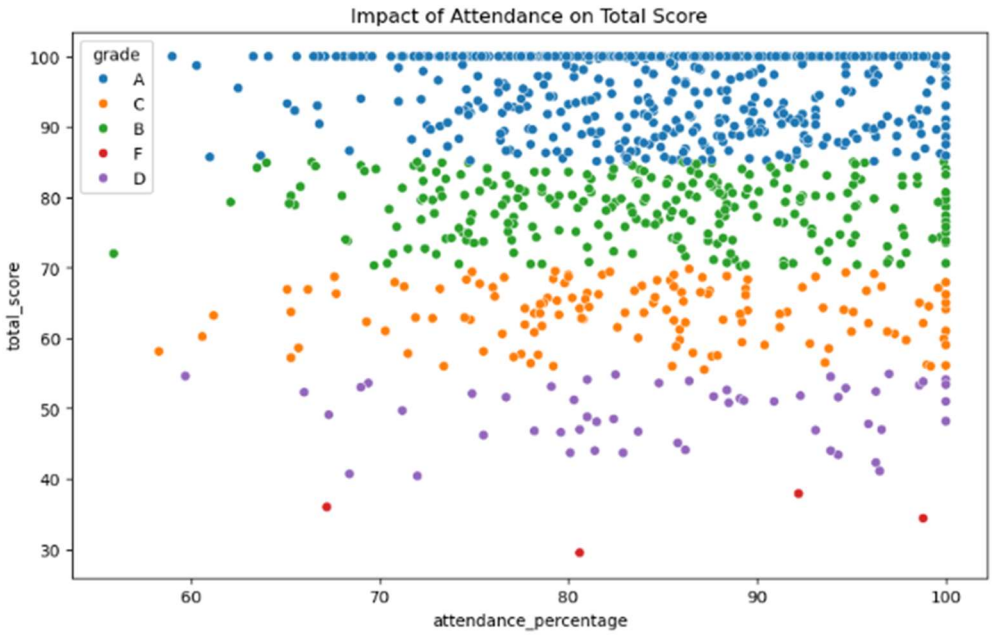
```
0         0
1         1
2         0
3         0
4         0
         ..
999995    0
999996    0
999997    0
999998    0
999999    0
Name: grade_num, Length: 1000000, dtype: int64
```

Step 3: EDA(Exploratory data analysis)

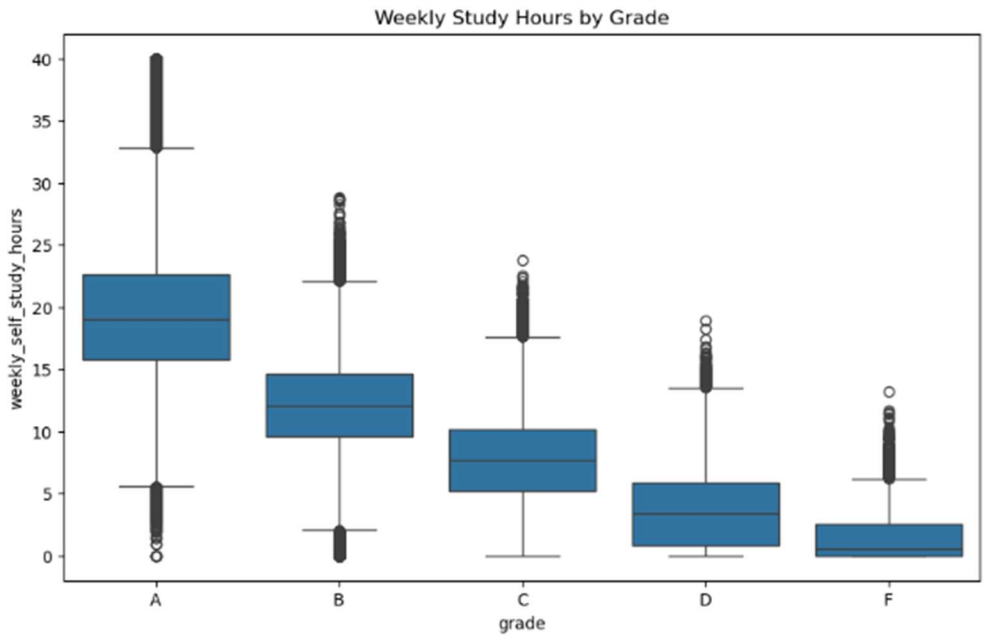
Exploratory Data analysis(EDA)

```
[7]: import matplotlib.pyplot as plt
import seaborn as sns
```

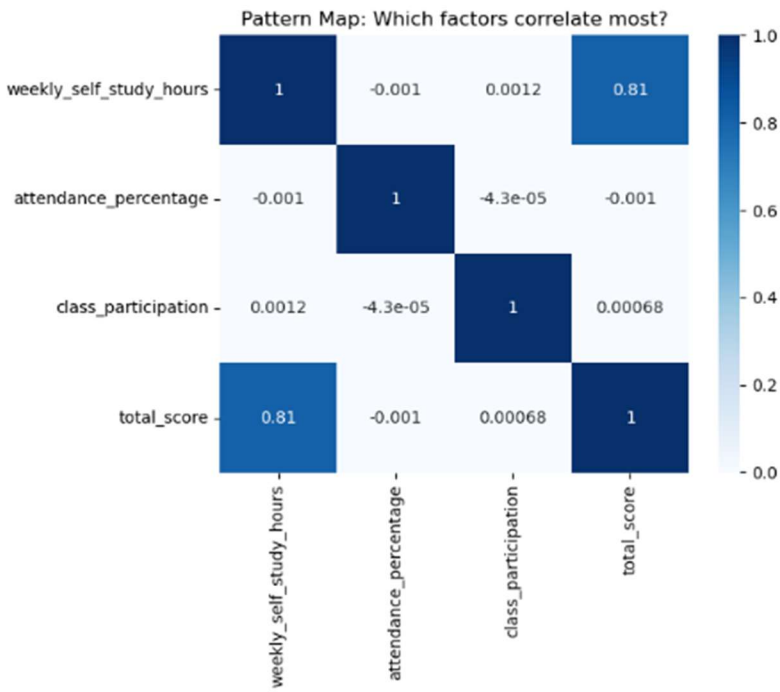
```
[8]: # Visualizing Attendance vs Total Score
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df.sample(1000), x='attendance_percentage', y='total_score', hue='grade')
plt.title('Impact of Attendance on Total Score')
plt.show()
```



[9]:
```
# Boxplot: Study Hours vs Grades
plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='grade', y='weekly_self_study_hours', order=['A', 'B', 'C', 'D', 'F'])
plt.title('Weekly Study Hours by Grade')
plt.show()
```



[10]:
```
numeric_columns = df[['weekly_self_study_hours', 'attendance_percentage', 'class_participation', 'total_score']]
correlation = numeric_columns.corr()
sns.heatmap(correlation, annot=True, cmap='Blues')
plt.title('Pattern Map: Which factors correlate most?')
plt.show()
```



Step 4: feature selection

Feature selection

```
[11]: X = df[['weekly_self_study_hours', 'attendance_percentage', 'class_participation']]
      y = df['grade_num']

[12]: from sklearn.model_selection import train_test_split

[13]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 5:Model training

Apply multiple ML algorithms:

- **Logistic Regression**
- **Decision Tree**
- **Naive Bayes**
- **Support Vector Machine (SVM)**

MODEL TRAINING MODULE

model selection and model training

[14]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
```

[15]:
```
#logistic regression
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)
```

```
C:\Users\HP\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py:465: ConvergenceWarning: lbfgs f
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

[15]:
```
LogisticRegression
LogisticRegression(max_iter=1000)
```

[16]:
```
# 2. Decision Tree
dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
```

[16]:
```
DecisionTreeClassifier
DecisionTreeClassifier()
```

[17]:
```
# 3. Naive Bayes
nb = GaussianNB()
nb.fit(X_train, y_train)
```

[17]:
```
GaussianNB
GaussianNB()
```

[18]:
```
svm = SVC(kernel='linear')
svm.fit(X_train[:5000], y_train[:5000])
```

[18]:
```
SVC
SVC(kernel='linear')
```

Step 7 & 8 Prediction and evaluation

PREDICTION MODULE

prediction & evaluation

[20]:
```
print("--- Model Accuracy Results ---")
print(f"Logistic Regression: {accuracy_score(y_test, lr.predict(X_test)):.2%}")
print(f"Decision Tree: {accuracy_score(y_test, dt.predict(X_test)):.2%}")
print(f"Naive Bayes:{accuracy_score(y_test, nb.predict(X_test)):.2%}")
print(f"SVM (Linear):{accuracy_score(y_test[:1000], svm.predict(X_test[:1000])):.2%}")
```

```
--- Model Accuracy Results ---
Logistic Regression: 69.69%
Decision Tree: 60.91%
Naive Bayes:69.65%
SVM (Linear):67.50%
```

```
[21]: final_predictions = lr.predict(X_test)
```

```
[22]: # 2. Create a clean results table
results_df = X_test.copy()
results_df['Predicted_Grade_Num'] = final_predictions
```

```
[23]: # Map numbers back to Letters for the report
reverse_map = {0: 'A', 1: 'B', 2: 'C', 3: 'D', 4: 'F'}
results_df['Predicted_Grade'] = results_df['Predicted_Grade_Num'].map(reverse_map)
```

```
[24]: at_risk_students = results_df[results_df['Predicted_Grade'].isin(['D', 'F'])]
```

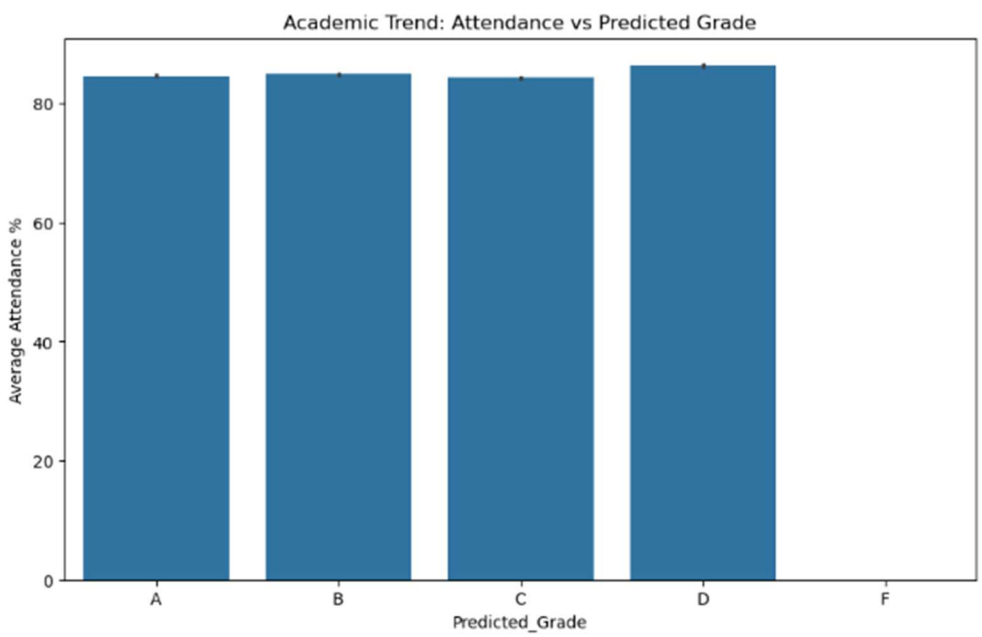
```
[27]: print(f"\nTotal Students Identified as 'At-Risk': {len(at_risk_students)}")
print("Top 5 people who need attention")
print(at_risk_students.head())
```

```
Total Students Identified as 'At-Risk': 6192
Top 5 people who need attention
        weekly_self_study_hours  attendance_percentage  class_participation  \
55399                       0.9                   87.0                  5.4
943378                      0.4                   84.8                  4.2
67898                       0.0                   69.4                  7.5
837523                      2.0                   86.2                 10.0
889145                      0.8                   82.6                  3.8

        Predicted_Grade_Num Predicted_Grade
55399                     3               D
943378                    3               D
67898                     3               D
837523                    3               D
889145                    3               D
```

RESULT VISUALIZATION MODULE

```
[26]: plt.figure(figsize=(10, 6))
sns.barplot(x='Predicted_Grade', y='attendance_percentage', data=results_df, order=['A', 'B', 'C', 'D', 'F'])
plt.title('Academic Trend: Attendance vs Predicted Grade')
plt.ylabel('Average Attendance %')
plt.show()
```



Output shows

• Student performance prediction

• Graphs showing trends

• List of at-risk students

Advantages

- Early detection of weak students
- Improves teaching strategies
- Data-driven decision making
- Saves time for teachers

Limitations

• Requires quality dataset

• Predictions depend on data accuracy

• Cannot capture emotional/psychological factors

Future Enhancements

• Integration with mobile app

• Real-time performance tracking

• AI-based personalized learning recommendations

• Integration with LMS (Learning Management Systems)

Conclusion

The system helps educational institutions:

• Predict student outcomes accurately

• Improve academic performance

• Use machine learning for smart education system